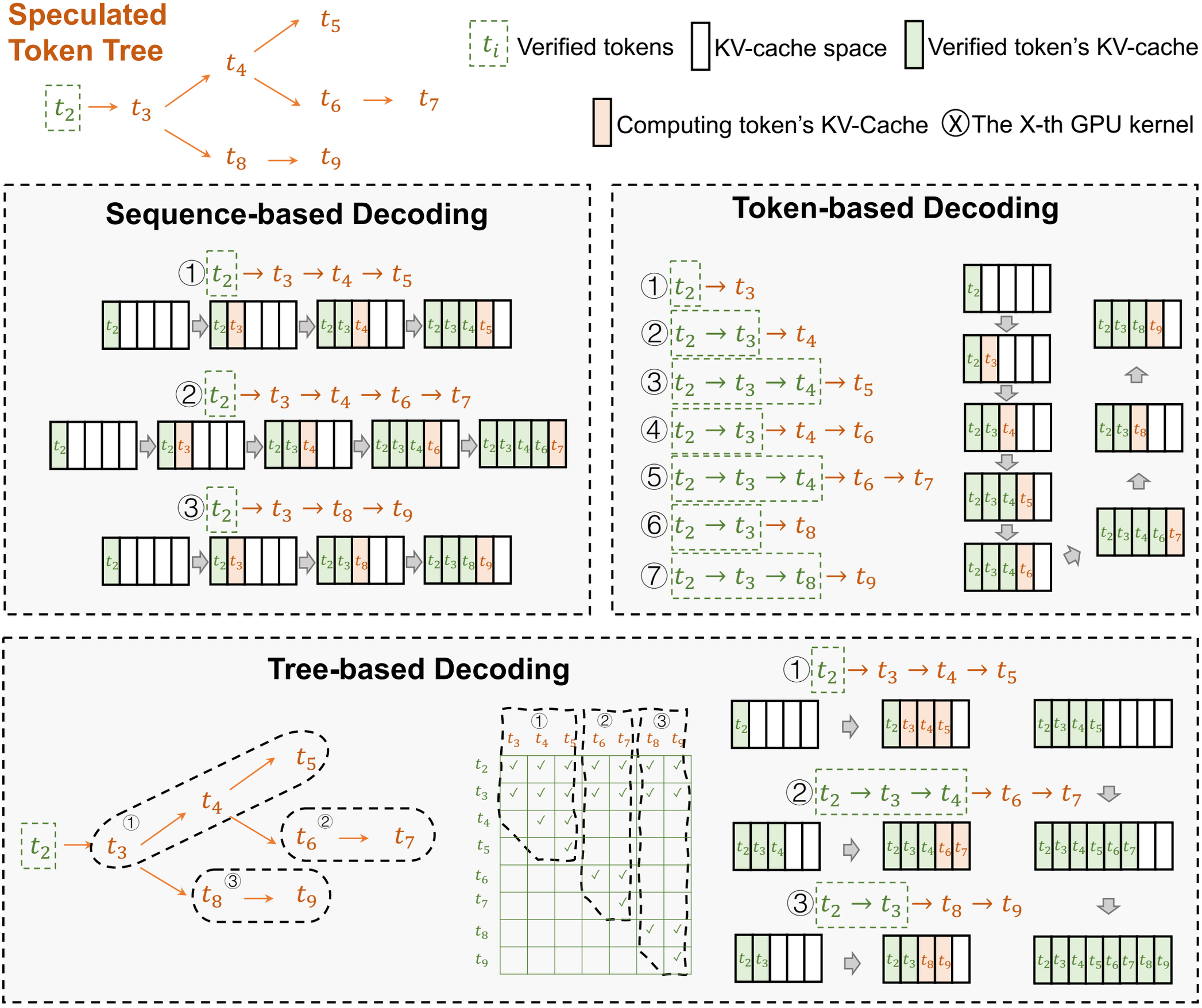
Speculated Token Tree
t_2 t_3 t_4 t_5 t_6 t_7 t_8 t_9
t_i Verified tokens
KV-cache space
Verified token's KV-cache
Computing token's KV-Cache
ⓧ The X-th GPU kernel
Sequence-based Decoding
① $t_2 \to t_3 \to t_4 \to t_5$
② $t_2 \to t_3 \to t_4 \to t_6 \to t_7$
③ $t_2 \to t_3 \to t_8 \to t_9$
Token-based Decoding
① $t_2 \to t_3$
② $t_2 \to t_3 \to t_4$
③ $t_2 \to t_3 \to t_4 \to t_5$
④ $t_2 \to t_3 \to t_4 \to t_6$
⑤ $t_2 \to t_3 \to t_4 \to t_6 \to t_7$
⑥ $t_2 \to t_3 \to t_8$
⑦ $t_2 \to t_3 \to t_8 \to t_9$
Tree-based Decoding
① $t_2 \to t_3 \to t_4 \to t_5$
② $t_2 \to t_3 \to t_4 \to t_6 \to t_7$
③ $t_2 \to t_3 \to t_8 \to t_9$